Notas de Combinatória 2

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March 28, 2026

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0 Informações importantes

Devo guardar notas de aula e exercícios interessantes do curso. Por enquanto nenhuma notação muito esotérica apareceu, se sim devo listá-las aqui. O objetivo dessas notas é serem breves, de forma que é possível recuperar formalmente as provas dos teoremas enunciados. Não serei estritamente rigoroso, o que pode levar a alguns erros. A linguagem pode variar intermintentemente entre português e inglês.

O código fonte desse arquivo pode ser encontrado em `https://github.com/hnrq104/comb2/tree/main`.

1 Extremal Set Theory

Lemma 1.1. (LYMB) Let $\mathcal{A} \subset \mathcal{P}([n])$ be an anti-chain, then

$$\sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1} \leqslant 1$$

Proof. (Sketch) Take a random permutation and consider the indicator random variables X_A that the first $|A|$ elements form A. We find that $\mathbb{E} \sum X_A = \sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1}$, but $\sum X_A \leqslant 1$. □

Corollary 1.2. (Sperner) If $\mathcal{A}$ is an anti-chain, $|\mathcal{A}| \leqslant \binom{n}{n/2}$.

Corollary 1.3. (Littlewood-Offord) Given $a_1, a_2, \ldots a_n \in \mathbb{R} \setminus \{0\}$. Consider $x_1, \ldots x_n \in \{-1, 1\}$ u.i.i.d r.v's, then $\mathbb{P}(\sum a_i x_i = 0) \leqslant 2^{-n} \binom{n}{n/2}$.

Proof. By simmetry of the x_i, we may assume that all a_i are positive. Now consider the collection $\mathcal{A} := \{A \subset [n] : \sum_{i \in A} a_i - \sum_{i \notin A} a_i = 0\}$. $\mathcal{A}$ is an anti-chain. $\square$

Theorem 1.4. (Erdos-Ko-Rado, Katona) Let $\mathcal{A} \subset \binom{[n]}{k}$ be an intersecting family with $2k < n$, then $|\mathcal{A}| \leqslant \binom{n-1}{k-1}$.

Proof. (Sketch) Take a random circular permutation of $[n]$, let X_A be the indicator of A appearing as an interval on the circle. We find that $\sum X_A \leqslant k$ - partition the intervals that intersect A and use pigeonhole. Calculating expectation and rearranging gives the result. $\square$

Theorem 1.5. (Exercise ERK) There is $n_0(k, l)$ such that if $n \geqslant n_0$ and $\mathcal{A} \subset \binom{[n]}{k}$ is an l-intersecting family, then $|\mathcal{A}| \leqslant \binom{n-l}{k-l}$. Hints:

1. Show that there exists $L \subset [n]$, $|L| = l$ such that $L = A \cap B$ for $A, B \in \mathcal{A}$.
2. Show that either $\mathcal{A} \subset \partial L := \{A \in \binom{[n]}{k} : L \subset A\}$ or $|\mathcal{A}| = O(n^{k-l-1})$.

Proof. For the first step, assume $\mathcal{A}$ is not empty and for every $A, B \in \mathcal{A}$, $|A \cap B| \geqslant l + 1$. Choose any element C from $\mathcal{A}$ and notice that every set in the family must intersect at least with $l + 1$ elements from C, so the number of sets in $\mathcal{A}$ is bounded by

$$|\mathcal{A}| \leqslant \binom{k}{l+1}\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

So we may assume that there are A, B, L as in the hint. Now, either $\mathcal{A} \subset \partial L$ and we are done, or there exists $C \in \mathcal{A}$ such that $L \not\subset C$. How many such C can exists? Because $A \cap B = L$, $L \not\subset C$, $|A \cap C| \geqslant l$ and $|B \cap C| \geqslant l$ we find that

$$|C \cap (A \cup B)| = |C \cap (A \cap B)| + |C \cap (A \setminus B)| + |C \cap (B \setminus A)| \geqslant l + 1$$

where we used the fact that $l \leqslant |C \cap A| = |C \cap L| + |C \cap (A \setminus B)|$ and so $|C \cap (A \setminus B)| \geqslant 1$. So the number of such C is at most $\binom{|A \cup B|}{l+1}\binom{n-|A \cup B|}{k-l-1} = O(n^{k-l-1})$. And now, fix any such C, how many elements of $\mathcal{A}$ may contain L? Because they all intersect in C an element outside of L, we have at most

$$|C \setminus L|\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

such elements, which completes the proof. $\square$

I really liked Rob's motivation for continuing to study these extremal problems.

(a) An explicit construction that gives $R(k) \geqslant k^C$ for any fixed C!

(b) The chromatic number of $\mathbb{R}^n$. We define $\chi(\mathbb{R}^n)$ as the minimal number of colors k such that there exists a function $f : \mathbb{R}^n \to [k]$ satisfying that for every $x, y \in \mathbb{R}^n$ with $|x - y| = 1$, we have that $f(x) \neq f(y)$. For example, we know by taking a simplex on $\mathbb{R}^n$ that $\chi(\mathbb{R}^n) \geqslant n+1$. It is possible to show (by considering a tesselation by cubes with diameter less than 1) that the number is well defined for any dimension. We will use extremal stuff to prove some order of growth in n of this parameter.

(c) (Borsuk's Conjecture) Show that for every set $X \subset \mathbb{R}^n$ it is possible to decompose it in $n+1$ sets $X_1, \dots X_{n+1}$ s.t. $\operatorname{diam}(X_i) < X$.

Now lets go back to some theorems. The next few ones are nice applications of linear algebra.

Definition 1.6. Given $L \subset [n]$, we say a family $\mathcal{A} \subset \mathcal{P}(n)$ is L-intersecting if for every $A \neq B \in \mathcal{A}$ we have that $|A \cap B| \in L$.

Theorem 1.7. Let $\mathcal{A} \subset \mathcal{P}(n)$ and let $L \subseteq \mathbb{Z}_p$ with $|L| = s$. If $\mathcal{A}$ is L-intersecting and for every $A \in \mathcal{A}$, $A \pmod p \notin L$, then

$$|\mathcal{A}| \leqslant \sum_{i=0}^{s} \binom{n}{i}$$

Proof. Consider the L.I set

$$f_A(X) = \prod_{l \in L} (X \cdot \mathbb{1}_A - l)$$

in $\mathbb{F}_p[x_1, x_2 \dots x_n]$ and calculate the dimension. □

Theorem 1.8. Let $\mathcal{A} \subset \binom{n}{k}$ be L intersecting, then $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$.

Proof. Consider $f_A = \prod_{l \in L}(X \cdot \mathbb{1}_A - l) \in \mathbb{R}[x_1, \dots, x_n]$, this always has degree less than $|L|$ and all monomials (after correcting the exponents) have degree at most $|L|$. So, by the dimension bound (of vectors being L.I), we know that $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$. □

Now let's sketch some applications.

Theorem 1.9. There is a explicit coloring that shows $R(m) \geqslant m^C$ for any constant C. (In our setup) There exists a sequence of graphs on N vertices with no clique or independent set of size $N^{o(1)}$.

Proof. The idea is to consider a graph, such that cliques and independent sets correspond to L intersecting families. So let's fix our $V(K_N) = \binom{[n]}{k}$. And consider a coloring of $E[K_N]$ by making

$$c(A, B) = \begin{cases} \text{R} & \text{if} \quad |A \cap B| \in L \\ \text{G} & \text{if} \quad |A \cap B| \notin L \end{cases}$$

where $L \subset [k-1]$. Now, what's the maximum size of a red clique? We know by [1.8], that it is at most by $\binom{n}{\leqslant |L|}$. So if L is relatively small, we can expect to win something. What about green cliques? Then we turn to the modular theorem [1.7]. If we suppose $k = p^2 - 1$ and let $L = \mathbb{Z}_p \setminus \{-1\}$. We would have that each green clique satisfies the hypothesis of 1.7, and so has size at most $\binom{n}{\leqslant p-1}$.

So thats the idea, we take $L = \{p-1, 2p-1, \dots p(p-1)-1\}$ so our coloring has no red clique of size $\binom{n}{\leqslant p-1}$, and no green clique of size $\binom{n}{\leqslant p-1}$.

Suffices to show that if $N = \binom{n}{k}$, then $\binom{n}{\leqslant \sqrt{k}} = N^{o(1)}$. □

Theorem 1.10. $\chi(\mathbb{R}^n) \geqslant \exp(cn)$

Proof. The proof follows similarly, we somehow want to find a finite embeddable graph in $\mathbb{R}^n$ with small independence number. Then $\chi(\mathbb{R}^n) \geqslant \chi(G) \geqslant |V(G)|/\alpha(G)$. So, let's look into our setting again of $V(G) = \binom{[n]}{k}$, and to embedd stuff in $\mathbb{R}^n$ we will consider the indicator functions k-sets in $\mathbb{R}^n$ (which are vectors of n dimensions).

Now we want to forbid a distance between these vectors (which will make our sets independent). So let's forbid a specific distance (which we can always scale to be the right thing) and say

$$E(G) = \{(A, B) : |A \cap B| \neq t\}$$

For some $t \in \mathbb{Z}$. Here as we don't really care about the clique number (only the independence), so we need to bound the size of a independent set. So in order to get the strongest bound, we use theorem [1.7] and suppose $k = 2p - 1$ for some p and $L = t \pmod p$. We get $\alpha(G) \leqslant \binom{n}{1}$. So by taking k very big (say $n/2$) we get the result.

□

Theorem 1.11. (Disproof of Borsuk's Conjecture) $\mathrm{Bor}(\mathbb{R}^n) \geqslant (1+\varepsilon)^{\sqrt{n}}$.

Proof. The proof can be found in Rob's book from 2010 with Imbuzeiro, but follows the same paradigm. □

2 Ramsey Theory

Just for sake of completeness, we define

$$R(k, l) = \min\{n : K_k \not\subset G \subset K_n \Rightarrow \alpha(G) \geqslant l\}.$$

In coloring terms, it's the smallest n such that any red and blue coloring of $E(K_n)$ contains a red K_k or a blue K_l.

2.1 Bounds on R(3,k)

We have two very simple bounds that are easy to prove,

$$\left(\frac{k}{\log(k)}\right)^{3/2} \leqslant R(3, k) \leqslant \binom{k+1}{2}.$$

The upperbound follows easily from the Erdös-Szekeres proof.

Theorem 2.1. $R(3, k) \leqslant \binom{k+1}{2}$

Proof. Let G on n-vtx be a triangle-free with no independet set of size k, we know every neighborhood is an independent set so $\Delta(G) < k$. Now we greedily get one vtx at a time removing their neighbors. Pick v_1 and $N(v_1)$ with $|N(v_1)| \leqslant k - 1$, then pick v_2, notice that the neighborhood of v_2 (after removing v_1 and its neighbors) has size at most $k - 2$, if not $\{v_1\} \cup N(v_2)$ would be a independent set of size k. We proceed like this for $v_1, v_2, \ldots v_t$, where each step we discard at most $k - (i - 1)$ vtxs. Because we can't do this process k times, for some $i < k$ we must fail (meaning there are no vertex left), so $n = |V(G)| \leqslant \sum_{j=0}^{k-2} k - j \leqslant \binom{k+1}{2} - 1$. □

The lowerbound is a beautiful application of the probabilistic method. The main idea is to choose $G(n,p)$ as to have few triangles (linearly small in n) and relatively low independence number.

Theorem 2.2. $R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^{3/2}$

We need a **very important** lemma first.

Lemma 2.3. With high probability, the independence number of $G(n,p) \leqslant 2\log(n)/p$. That is

$$\mathbb{P}(\alpha(G(n,p)) \geqslant 2\log(n)/p + 1) = 1 - o_{(n,p(n))}(1)$$

where p can be taken polynomially small in n.

Proof.

$$\begin{aligned}\mathbb{P}(\alpha(G(n,p)) \geqslant k) &\leqslant \mathbb{E}[\#\text{independent k sets} \in G(n,p)] \\ &= \binom{n}{k}(1-p)^{\binom{k}{2}} \leqslant \left(\frac{en}{k}\right)^k \cdot e^{-pk(k-1)/2} \\ &= \left(\frac{en}{e^{p(k-1)/2}k}\right)^k\end{aligned}$$

Taking $k \geqslant 2\log(n)/p + 1$ we get that

$$\mathbb{P}(\alpha(G(n,p)) \geqslant k) \leqslant \left(\frac{ep}{2\log(n)}\right)^{2\log(n)/p} \to 0$$

very fast even! (faster than any polynomial). □

The strategy will be to remove triangles from $G(n,p)$ by removing one vertex from each. This cannot increase the independence number, so we hope the resulting graph still has many vertices to yield a Ramsey construction.

So let's see what this alteration may give us.

Proof. (Of the Theorem) We choose p small such that the number of triangles is no larger than a small linear in n (say $n/2$).

$$\mathbb{E}[\#K_3 \subset G(n,p)] = \binom{n}{3}p^3 = (1+o(1))\frac{(pn)^3}{6}$$

We want

$$\mathbb{P}(\#K_3 \in G(n,p) \geqslant n/2) < (1+o(1))\frac{p^3n^2}{3} < 1/2$$

so taking $p = \varepsilon n^{-2/3}$ for some small ε we get the inequality. But what does this p gives in independence number? We have w.h.p (say bigger than one half)

$$\alpha(G(n,p)) \leqslant 2\varepsilon^{-1}\log(n)n^{2/3}.$$

So $\mathbb{P}(K_3(G(n,p)) < n/2) > 1/2$ and $\mathbb{P}(\alpha(G(n,p) < C\log(n)n^{2/3})) > 1/2$, meaning there exists G satisfying both statements. Remove one vtx from each triangle in G, we are left with a triangle-free graph G' with at least $n/2$ vtxs and $\alpha(G) < C\log(n)n^{2/3}$. Setting $k = C\log(n)n^{2/3}$, we get that

$$V(G') = n/2 \geqslant c\left(\frac{k}{\log(k)}\right)^{3/2}$$

□

Now for a better (and astounishingly beautiful) bound.

Theorem 2.4.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^2.$$

The idea of this bound is removing edges. Maybe we can kill the triangles by removing few edges and not increase the independence number too much. For this we will need a beautiful lemma.

Lemma 2.5. (Erdös-Tetalli) Let $\{A_1, A_2, \dots A_m\}$ be a family of events in some probability space (Ω, P). Let $\mathcal{E}_t$ be the event that at least t different independent A_i happen. We have that

$$\mathbb{P}(\mathcal{E}_t) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

Proof. We get this one from just opening the terms

$$\mathbb{P}(\mathcal{E}_t) = \mathbb{P}\left(\bigcup_{\substack{i_1<\dots<i_t \\ A_{i_j}\not\sim A_{i_k}}} \bigcap_{j=1}^{t} A_{i_j}\right) \leqslant \sum_{\substack{i_1<\dots<i_t \\ A_{i_j}\not\sim A_{i_k}}} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \sum_{i_1<\dots<i_t} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

□

The second ingredient is noting that if you're a bit bigger (say a constant times) than the independent size, you already have many edges (at least half of what is expected).

Lemma 2.6. Let $k \geqslant C\log(n)/p$, then

$$\mathbb{P}\left(\exists \text{ k-set } S, e(G(n,p)[S]) < p\binom{k}{2}/2\right) = o(1)$$

for some $C > 0$.

Proof. Let $\mathcal{E}$ be the event that there exists a k-set S with $e(G(n,p)[S]) < p\binom{k}{2}/2$. We will do a union bound and use Chernoff to bound its probability.

Let S be any k-set. Note that (letting $B(n,p)$ be the binomial with parameters n and p) we have

$$e(G(n,p)[S]) \equiv B\left(\binom{k}{2}, p\right).$$

So by Chernoff (or Hoeffding) [**??**] (need to add proofs for these inequalities),

$$\mathbb{P}\left(e(G(n,p)) < \frac{p}{2}\binom{k}{2}\right) \leqslant \mathbb{P}\left(\left|B\left(\binom{k}{2}, p\right) - p\binom{k}{2}\right| \geqslant \frac{p}{2}\binom{k}{2}\right)$$
$$\leqslant e^{-cp\binom{k}{2}}$$

for some $c > 0$.

Now, doing the union bound over k-sets we find that (taking $C > 3c^{-1}$)

$$\mathbb{P}(\mathcal{E}) \leqslant \binom{n}{k} e^{-cp\binom{k}{2}} \leqslant \left(\frac{en}{e^{cp(k-1)/2}k}\right)^k \leqslant \left(\frac{e}{k}\right)^k \to 0.$$

□

The idea to prove theorem 2.4 is to remove a maximal collection of edge-disjoint triangles from $G(n,p)$, (notice that this makes it triangle free). We use lemma 2.5 to show that this procedure won't remove too many edges from any set S with size $C\log(n)/p$.

Proof. (Of Theorem 2.4) Let S be a set of size $k = C\log(n)/p$, for some p we will determine later. Let $\{T_1, \dots T_m\}$ be the set of all triangles that intersect with at least a pair in S. Note that $m \leqslant n\binom{|S|}{2}$. Let $\mathcal{E}_S$ be the event that a edge disjoint set of these triangles with size bigger or equal to $\frac{p}{6}\binom{k}{2}$ are in $G(n,p)$. By [2.5], we can bound $\mathbb{P}(\mathcal{E}_S)$,

$$\mathbb{P}(\mathcal{E}_S) \leqslant \frac{1}{[\frac{p}{6}\binom{k}{2}]!}\left(\sum_{i=1}^{m} \mathbb{P}(T_i \in G(n,p))\right)^t \leqslant \left(\frac{24e}{pk^2}\right)^{pk^2/24} (p^3nk^2)^{pk^2/12}$$

Replacing k in the equation we get (and putting $C > 24e$)

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{24ep}{C\log^2(n)}\right)^{\log(n)^2/(24p)} (pn\log(n)^2)^{\log(n)^2/(12p)} \leqslant (p^3n^2\log(n)^2)^{\log(n)^2/(24p)}$$

So, if we take $p = \varepsilon n^{-1/2}$ (might try to optimize more later), we already find that

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{\log(n)}{n^{1/2}}\right)^{n^{1/2}\log^2(n)} \ll \left(\frac{C\log(n)}{n^{1/2}}\right)^{C\log(n)n^{1/2}} \leqslant \binom{n}{k}^{-1}$$

So $\mathcal{E}_S$ does not happen for any S with high probability. As w.h.p S of size k have more than $\frac{p}{2}\binom{k}{2}$ edges, and any maximal disjoint triangle family intersect too much with S. We can remove a maximal edge disjoint triangle family and not make any set of size k independent. This yields a triangle free graph on n vtxs with independence number less than $C\log(n)n^{1/2}$, so

$$R(3,k) \geqslant c\left(\frac{k}{\log k}\right)^2.$$

□